

Model Lab

Every prior, same slate — 2026-09-19

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

36,473

Held-Out Scores

Fitted on plate appearances before 2026-08-13, scored on the 36,473 that came after (actual HR rate 2.97%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5708	1.44×	0.02879	0.13298	0.00333
KNN comps Prior is the pooled rate of the nearest comparables.	0.5906	1.51×	0.02878	0.13248	0.00428
SVR Support-vector regression on the same feature matrix.	0.5797	1.44×	0.02885	0.13405	0.00779
Random forest Forest regression; a learned similarity metric.	0.5838	1.48×	0.02884	0.13357	0.00713
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5906	1.51×	0.02878	0.13248	0.00428
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5906	1.51×	0.02878	0.13248	0.00428
Linear regression Ordinary least squares on the same features.	0.5882	1.45×	0.02883	0.13347	0.00623
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5896	1.46×	0.02884	0.13308	0.00715
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5899	1.42×	0.02882	0.13300	0.00630
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5816	1.49×	0.02888	0.13395	0.00779

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 244, knn 244, svr 244, rf 244, core 244, ensemble 244, linear 244, logistic 244, nn 244, xgb 244 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,882 PA, actual HR rate 2.50%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5060	1.03×	0.02439	0.11739	0.00877
KNN comps Prior is the pooled rate of the nearest comparables.	0.5772	1.34×	0.02445	0.11794	0.01126
SVR Support-vector regression on the same feature matrix.	0.5575	1.24×	0.02478	0.12544	0.02143
Random forest Forest regression; a learned similarity metric.	0.5625	0.93×	0.02480	0.12366	0.02240
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5772	1.34×	0.02445	0.11794	0.01126
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5772	1.34×	0.02445	0.11794	0.01126
Linear regression Ordinary least squares on the same features.	0.5681	0.83×	0.02488	0.12494	0.02187
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5761	0.83×	0.02484	0.12169	0.02228
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5788	1.24×	0.02474	0.12102	0.01827
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5647	0.83×	0.02514	0.12596	0.02592

Slate Agreement

Flat league · 4/6 shared 0/5 hit

Pete Crow-Armstrong

Junior Caminero

Willson Contreras

Francisco Lindor

Elly De La Cruz

Ben Rice

SVR · 5/6 shared 0/5 hit

Pete Crow-Armstrong

Junior Caminero

Emmanuel Rodriguez

Juan Soto

Ben Rice

Willson Contreras

Core (KNN+SVR) · 6/6 shared 0/5 hit

Pete Crow-Armstrong

Willson Contreras

Junior Caminero

Munetaka Murakami

Juan Soto

Ben Rice

Linear regression · 3/6 shared 0/5 hit

Pete Crow-Armstrong

Emmanuel Rodriguez

Willson Contreras

Lazaro Montes

Junior Caminero

Kyle Schwarber

Neural net (torch) · 4/6 shared 0/5 hit

KNN comps · 6/6 shared 0/5 hit

17.2% Pete Crow-Armstrong

15.8% Willson Contreras

14.5% Junior Caminero

14.4% Munetaka Murakami

14.3% Juan Soto

14.2% Ben Rice

Random forest · 4/6 shared 0/5 hit

20.5% Pete Crow-Armstrong

19.1% Emmanuel Rodriguez

18.9% Lazaro Montes

17.6% Willson Contreras

17.5% Munetaka Murakami

17.3% Ben Rice

Forest blend · 6/6 shared 0/5 hit

19.3% Pete Crow-Armstrong

17.6% Willson Contreras

16.9% Junior Caminero

16.4% Munetaka Murakami

16.4% Juan Soto

16.4% Ben Rice

Logistic regression · 2/6 shared 0/5 hit

19.4% Emmanuel Rodriguez

19.0% Pete Crow-Armstrong

17.9% Willson Contreras

17.5% Lazaro Montes

16.8% Kyle Schwarber

16.6% Yordan Alvarez

19.3%

17.6%

16.9%

16.4%

16.4%

16.4%

20.1%

19.8%

18.3%

17.5%

17.1%

17.1%

19.3%

17.6%

16.9%

16.4%

16.4%

16.4%

21.9%

20.9%

19.5%

18.5%

18.4%

17.6%

Ben Rice

16.9%

Willson Contreras

17.5%

Hitter by Hitter

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
1. Pete Crow-Armstrong slate Chicago Cubs · vs LHP · 44 HR / 692 PA	17.2%	19.3%	20.5%	20.1%	19.3%	19.3%	19.4%	20.9%	19.9%	20.2%
2. Willson Contreras slate Boston Red Sox · vs RHP · 27 HR / 534 PA	14.5%	17.6%	17.3%	17.5%	17.6%	17.6%	17.9%	19.5%	17.9%	18.2%
3. Junior Caminero Tampa Bay Rays · vs RHP · 41 HR / 669 PA	15.8%	16.9%	19.1%	16.8%	16.9%	16.9%	16.8%	17.5%	17.6%	17.8%
4. Munetaka Murakami slate Chicago White Sox · vs RHP · 32 HR / 517 PA	13.9%	16.4%	17.0%	17.1%	16.4%	16.4%	16.2%	16.4%	16.5%	16.7%
5. Juan Soto slate New York Mets · vs RHP · 26 HR / 444 PA	13.4%	16.4%	17.6%	16.9%	16.4%	16.4%	16.3%	16.6%	16.8%	17.0%
6. Ben Rice slate New York Yankees · vs RHP · 39 HR / 633 PA	14.2%	16.4%	17.5%	17.1%	16.4%	16.4%	16.5%	16.8%	16.9%	17.1%
7. Emmanuel Rodriguez slate Minnesota Twins · vs RHP · 1 HR / 28 PA	10.9%	16.0%	18.9%	19.8%	16.0%	16.0%	19.0%	21.9%	19.7%	20.0%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
8. Pete Alonso Baltimore Orioles · vs LHP · 38 HR / 670 PA	13.2%	16.0%	16.2%	16.0%	16.0%	16.0%	15.9%	16.4%	16.0%	16.1%
9. Oneil Cruz Pittsburgh Pirates · vs RHP · 19 HR / 398 PA	13.6%	15.9%	15.9%	15.0%	15.9%	15.9%	15.3%	15.3%	15.7%	15.8%
10. Yordan Alvarez Houston Astros · vs RHP · 40 HR / 654 PA	13.4%	15.8%	17.2%	16.6%	15.8%	15.8%	16.2%	17.6%	16.8%	16.9%
11. Brandon Lowe Pittsburgh Pirates · vs RHP · 32 HR / 625 PA	14.2%	15.8%	16.3%	16.3%	15.8%	15.8%	16.0%	16.9%	16.3%	16.4%
12. Kyle Schwarber Philadelphia Phillies · vs RHP · 45 HR / 664 PA	13.7%	15.8%	17.1%	16.8%	15.8%	15.8%	16.6%	18.4%	16.7%	16.8%
13. Matt Olson Atlanta Braves · vs RHP · 40 HR / 667 PA	13.6%	15.6%	16.9%	16.4%	15.6%	15.6%	15.9%	17.2%	16.5%	16.6%
14. Roman Anthony Boston Red Sox · vs RHP · 5 HR / 215 PA	14.0%	15.5%	14.4%	16.5%	15.5%	15.5%	15.4%	14.3%	15.5%	15.6%
15. Wilyer Abreu Boston Red Sox · vs RHP · 24 HR / 653 PA	13.8%	15.4%	15.6%	15.7%	15.4%	15.4%	15.1%	15.6%	15.5%	15.6%
16. Miguel Vargas Chicago White Sox · vs RHP · 32 HR / 671 PA	12.9%	15.4%	15.2%	14.7%	15.4%	15.4%	15.3%	15.7%	15.6%	15.7%
17. Elly De La Cruz Cincinnati Reds · vs LHP · 28 HR / 601 PA	14.3%	15.3%	16.9%	16.0%	15.3%	15.3%	15.4%	16.0%	16.0%	16.1%
18. Lazaro Montes Seattle Mariners · vs RHP · 4 HR / 35 PA	11.4%	15.3%	12.5%	18.3%	15.3%	15.3%	17.5%	18.5%	17.5%	17.6%
19. Corey Seager Texas Rangers · vs RHP · 18 HR / 403 PA	12.2%	15.2%	15.1%	15.2%	15.2%	15.2%	14.9%	15.2%	15.5%	15.6%
20. Kazuma Okamoto Toronto Blue Jays · vs RHP · 33 HR / 604 PA	13.4%	15.2%	16.2%	15.4%	15.2%	15.2%	15.2%	16.1%	15.6%	15.7%
21. Francisco Lindor New York Mets · vs RHP · 20 HR / 430 PA	14.4%	15.2%	15.3%	14.9%	15.2%	15.2%	14.8%	14.9%	14.8%	14.9%
22. Colson Montgomery	13.0%	15.1%	15.2%	15.4%	15.1%	15.1%	15.9%	17.2%	15.8%	15.9%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	X
37. Dominic Canzone Seattle Mariners · vs RHP · 22 HR / 497 PA	12.3%	14.1%	14.5%	14.7%	14.1%	14.1%	14.2%	14.7%	14.7%	
38. Jackson Merrill San Diego Padres · vs RHP · 26 HR / 636 PA	11.9%	14.0%	13.8%	14.0%	14.0%	14.0%	14.1%	14.7%	14.3%	
39. Jonathan Aranda Tampa Bay Rays · vs RHP · 21 HR / 644 PA	12.3%	14.0%	13.1%	13.4%	14.0%	14.0%	13.4%	13.0%	13.1%	
40. Adley Rutschman Boston Red Sox · vs RHP · 12 HR / 383 PA	13.4%	13.9%	13.7%	12.7%	13.9%	13.9%	13.7%	13.1%	13.8%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.